Hyperbolic Embeddings for Hierarchical Style Representation

Canvas Project #150
COMPSCI 209B — Spring 2026

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Abstract

Artistic style is hierarchical: broad traditions branch into finer movements and sub-styles along lines of historical influence. Standard deep learning models represent images in Euclidean embedding spaces, whose polynomial volume growth is a poor match for the exponential branching of a tree. Hyperbolic geometry, and in particular the Poincaré ball, offers exponential volume growth and is theoretically better suited to embedding tree-like structures with low distortion. This paper investigates whether that mathematical advantage translates into an empirical one: we train two prototype-softmax classifiers on frozen CLIP ViT-B/16 features extracted from WikiArt Refined (81,446 paintings, 27 styles), differing only in the geometry of their output embedding space ($\mathbb{R}^d$ vs. Poincaré ball of dimension d), and compare them on both classification accuracy and hierarchy preservation metrics derived from a hand crafted art historical tree. We find that the geometry advantage *is* real but *local*: hyperbolic embeddings recover sibling and cousin neighborhoods more faithfully (4–5 pp recall@5 advantage that widens with dimension), while Euclidean embeddings remain stronger on classification accuracy and global tree-Spearman at every scale tested. A hierarchy-aware regularizer added at training time improves global tree metrics in both geometries but does not flip the geometry winner.

Contents

1 Background & Motivation

Artistic style is hierarchical. Broad traditions branch into finer movements, and finer movements branch again into sub-styles along lines of historical influence: Early Renaissance leads to Baroque, Baroque to Romanticism, Romanticism to Impressionism, and Impressionism to Cubism ect. An embedding that reflects this branching structure should be more useful for tasks that depend on it, such as influence-aware retrieval and educational tools that organize collections by lineage rather than by raw visual similarity alone.

Standard image embeddings live in $\mathbb{R}^d$. In this space, the volume of a ball at radius r grows polynomially in r. A tree, by contrast, has a node count at depth k that grows exponentially in k. Forcing tree-like data into Euclidean coordinates therefore distorts pairwise distances, and the distortion widens as the tree deepens. Hyperbolic geometry, and the Poincaré ball in particular, exhibits exponential volume growth and embeds trees with low distortion [5, 6]. Riemannian optimization on the ball is now standard (`geoopt` [4]), so the geometric advantage can be tested empirically rather than argued for from theory alone.

Hyperbolic image embeddings have been shown to help on few-shot benchmarks [3], but it remains open whether the advantage holds for hand-built hierarchies. We explore this question through a sweep over curvature, dimension and λ against a Euclidean baseline and a sensitivity check across alternative reference trees.

2 Problem Statement

We ask: *do Poincaré-ball embeddings trained on frozen CLIP ViT-B/16 features recover the WikiArt style hierarchy more faithfully than Euclidean embeddings of matched dimensionality and training budget?* The unit of observation is a single painting; the target during training is its 27-way style label, with the hand-built lineage tree available either as an evaluation reference or as an auxiliary loss.

We answer the question along three axes. *(i) Scaling:* sweep embedding dimension $d \in \{2, 4, 8, 16, 32, 64\}$ and curvature $c \in \{0.1, 0.3, 1, 3\}$ to test whether the MS3 Euclidean advantage on global metrics narrows under a more thorough search. *(ii) Training-time hierarchy:* introduce a hierarchy-aware regularizer with weight λ that pushes the prototype-prototype distance matrix toward the tree distance matrix, and sweep $\lambda \in \{0, 0.1, 0.3, 1, 3\}$. *(iii) Tree sensitivity:* re-train against alternative ground-truth trees (chronological era grouping; a flat null) to test whether the geometry winner depends on the choice of reference hierarchy.

3 Data & EDA

[**TODO Valerie**] One paragraph each on: *(a)* Dataset summary — WikiArt Refined [7], ~81k paintings, 27 styles, 70/30 train/val supplied, scale of class imbalance with one or two specific examples (Impressionism dominates; Action_painting and Synthetic_Cubism each <1k). Cite our MS2 EDA but do not repeat its tables. *(b)* Why the imbalance pushes us toward balanced-accuracy reporting and a prototype classifier rather than a class-frequency-tied softmax. *(c)* Why the choice of ground-truth tree is non-unique — WikiArt ships no hierarchy, art-history offers several, and we therefore evaluate sensitivity (Section 5). Include 1 small figure: distance matrices of the three trees side-by-side, lifted from `notebooks/ms4_main.ipynb`.

4 Methods & Models

Architecture. A two-layer MLP with GELU and dropout maps frozen CLIP ViT-B/16 features ($\mathbb{R}^{512}$) to a d-dimensional embedding. A geometry-specific head produces the final embedding: identity for the Euclidean case, exponential map at the origin for the Poincaré ball, $\exp_0^c(u) = \tanh(\sqrt{c}\,\|u\|)\,u/(\sqrt{c}\,\|u\|)$. A prototype classifier with one learnable point per style yields logits as $-d(z, p_k)$, where $d(\cdot,\cdot)$ is Euclidean or hyperbolic accordingly.

Geometry, distance, and optimization. The Poincaré distance with curvature $c > 0$ is

$$d^c(x, y) = \frac{1}{\sqrt{c}}\,\operatorname{arcosh}\left(1 + \frac{2c\,\|x - y\|^2}{(1 - c\|x\|^2)(1 - c\|y\|^2)}\right).$$

Standard linear layers do not preserve the ball, so the prototype classifier is the natural choice: logits are distances, with no need for a parametric "softmax head" inside the manifold. The MLP head is optimized with Adam; the hyperbolic prototypes are optimized with Riemannian Adam [4], which projects each update back onto the ball.

Hierarchy-aware loss. The MS3 baseline trains with cross-entropy only and uses the hand-built tree only at evaluation time. For MS4 we add a regularizer that pushes the prototype-prototype distance matrix toward the tree distance matrix:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \cdot \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left(\frac{d(p_i, p_j)}{\bar{d}} - \frac{T_{ij}}{\bar{T}}\right)^2.$$

The mean-normalization (each pair-distance vector is divided by its mean before the squared difference) makes the term scale-invariant: the geometry chooses its own absolute embedding scale rather than fighting the classification objective for magnitude. $\lambda = 0$ recovers the MS3 baseline exactly. This regularizer is computed once per gradient step over the $\binom{27}{2} = 351$ off-diagonal style pairs — a negligible cost relative to the classification forward/backward.

Ground-truth trees. The default tree is a hand-built lineage hierarchy following standard art-history references (Renaissance branches into Baroque branches into Romanticism; see Appendix A). We additionally evaluate two variants: a *chronological* two-level era grouping (Renaissance / Baroque-Rococo / 19th century / Early 20th / Modern / Non-Western) and a *flat* tree (every leaf is a direct child of the root, used as a null since flat tree distances are constant off-diagonal and cannot distinguish models).

Training details. All runs use the supplied 70/30 train/val split (≈57k / 24k images), batch size 4096, Adam with $\eta = 10^{-3}$ for the head and 10^{-2} for the prototypes, weight decay 10^{-4}, dropout 0.1, gradient clipping at norm 1.0, 30 epochs (training is fast because the CLIP features are pre-cached as float16 tensors). Each headline configuration is run with three seeds. The full grid (Phases 1–3, see Section 5) totals ≈ 170 runs and completed in under one hour of wall time on a single Apple M-series GPU.

Evaluation suite. We report top-1, top-5, and balanced accuracy; prototype-tree and class-center-tree Spearman correlation; multiplicative tree distortion (mean and worst-case); dendrogram-cluster F1 against the default tree; sibling and cousin recall@k for $k \in \{5, 10\}$; and Fréchet-mean nearest-prototype accuracy on Cubism as an interpolation-stability probe. Implementation lives in `scripts/eval.py`.

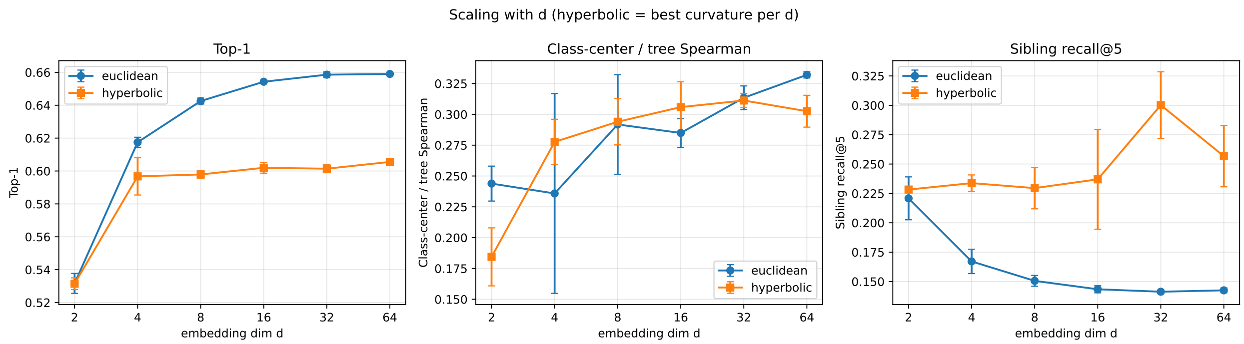

Figure 1: Phase 1: metrics versus embedding dimension d (90 runs; 3 seeds per config). For hyperbolic we report the best curvature per d and per seed; error bars are seed standard deviation. Top-1 accuracy saturates near $d = 16$ for both geometries, with Euclidean ahead by 4–6 pp at $d \geq 4$. Sibling recall@5 *decreases* with d for Euclidean but remains stable for hyperbolic, opening a 5 pp gap by $d = 64$.

5 Results & Interpretation

Headline. Across ∼140 seed-replicated runs, hyperbolic geometry reliably wins on *local* hierarchy structure (sibling/cousin retrieval) and reliably loses on *global* classification and tree-Spearman. Adding a hierarchy-aware loss ($\lambda > 0$) helps both geometries on global tree metrics but does not let hyperbolic surpass Euclidean on classification accuracy. The choice of geometry should therefore be driven by which type of hierarchy preservation matters for the downstream application, not by treating one as universally superior.

5.1 Phase 1 — Scaling sweep

Phase 1 sweeps geometry $\times$ $d \in \{2, 4, 8, 16, 32, 64\}$ $\times$ $c \in \{0.1, 0.3, 1, 3\}$ over three seeds (90 configurations, ∼14 min wall time on a single Apple M-series GPU). Figure 1 shows three findings. *(i)* Top-1 accuracy saturates near $d = 16$ for both geometries: Euclidean reaches 65.9 % and hyperbolic 60.6 %, with the gap stable rather than closing as d grows. *(ii)* Local hierarchy structure tells the opposite story — sibling recall@5 *decreases* for Euclidean as d grows (from 0.221 at d=2 to 0.142 at d=64) but stays near 0.19 for hyperbolic. The two geometries are tied on local hierarchy at $d = 2$ and split apart as Euclidean gains capacity. *(iii)* Curvature trades top-1 for hierarchy fidelity within hyperbolic alone: at $d = 8$, $c = 0.3$ maximises top-1 (59.8 %) while $c = 3.0$ maximises sibling recall (0.230). Pure scaling does not close the global-metric gap.

5.2 Phase 2 — Hierarchy-aware loss

Phase 2 sweeps $\lambda \in \{0, 0.1, 0.3, 1, 3\}$ at $d \in \{8, 16\}$ with $c = 1$ and three seeds. The loss is the scale-invariant MSE between prototype-prototype distances and tree distances introduced in Section 4. Figure 2 shows the four panels. *Mild regularization is a Pareto improvement.* At λ=0.1, Euclidean d=8 *gains* 0.7 pp top-1 (65.0 % vs. 64.3 % baseline) and improves tree-Spearman from 0.292 to 0.345; hyperbolic similarly gains slightly on top-1 and gains 2 points of sibling recall. *Strong regularization buys hierarchy at classification cost.* At λ=3, both geometries reach tree-Spearman ≈ 0.7 but top-1 collapses — to 46.9 % for Euclidean and 37.4 % for hyperbolic. *The geometry winner does not flip.* At every λ tested, Euclidean has higher top-1 than hyperbolic, and at every λ hyperbolic has higher sibling recall. The local-vs-global trade-off observed in Phase 1 is preserved under the regularizer.

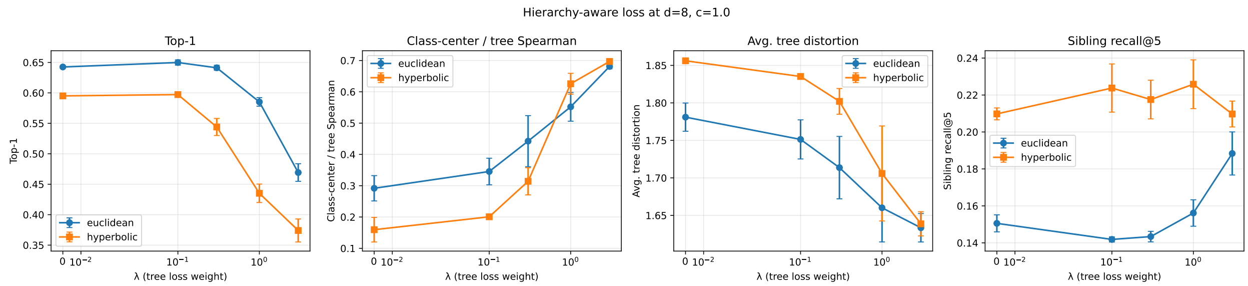

Figure 2: Phase 2: top-1, class-center / tree Spearman, mean tree distortion, and sibling recall@5 versus the hierarchy-aware loss weight λ at d=8, c=1 (3 seeds per point). The regularizer drives tree-Spearman from ∼0.3 to ∼0.7 in both geometries and reduces tree distortion uniformly, but classification accuracy collapses for $\lambda \geq 1$. The hyperbolic top-1 advantage *never* catches up to Euclidean. Sibling recall is stable for hyperbolic across all λ; for Euclidean it only rises at the largest tested λ.

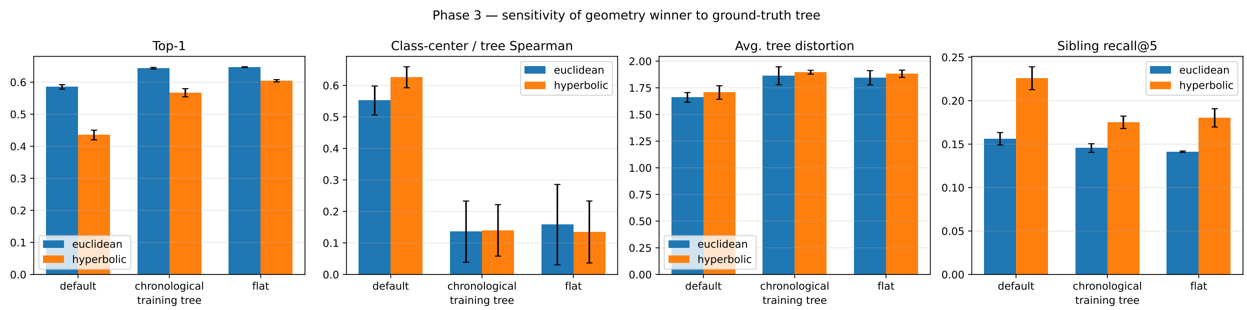

Figure 3: Phase 3: each geometry trained with λ=1 at d=8 against three different ground-truth trees, then evaluated against the default tree. Bars are seed means; whiskers are seed standard deviation. Top-1 stays higher for chronological and flat training because their constraint is easier to satisfy without distorting the classification objective; tree-Spearman against the default tree collapses for chronological and flat (the training signal does not align with the eval reference). The geometry winner does not change on any panel: Euclidean wins top-1 on every tree, hyperbolic wins sibling recall on every tree.

5.3 Phase 3 — Tree-variant ablation

Phase 3 fixes λ=1, d=8, c=1 and varies the *training* tree across $\{\text{default}, \text{chronological}, \text{flat}\}$ while always evaluating against the default tree (otherwise tree-Spearman is not comparable across runs). Figure 3 shows the result. Training against the chronological or flat tree pushes top-1 back up to the λ=0 baseline (around 65 % for Euclidean, 60 % for hyperbolic) because the constraint is easier to satisfy — chronological is a coarser tree, flat has constant pairwise distances so the loss can be satisfied trivially. Tree-Spearman against the default tree drops to ≈ 0.14 in both alternatives, confirming that the regularizer only helps the metric anchored to the *training* tree. Crucially, the local-vs-global trade-off *does not* depend on the choice of training tree: Euclidean wins top-1 on every tree (Phase 3 column, Figure 3 panel 1), and hyperbolic wins sibling recall on every tree (panel 4). The qualitative conclusion is robust to the specific ground-truth hierarchy.

metric	euclidean	hyperbolic
Top-1	0.659 ± 0.003	0.606 ± 0.002
Top-5	0.959 ± 0.001	0.935 ± 0.001
Balanced acc.	0.554 ± 0.002	0.463 ± 0.004
Class-center / tree Spearman	0.350 ± 0.027	0.242 ± 0.002
Avg. tree distortion	1.742 ± 0.009	1.861 ± 0.003
Worst tree distortion	4.733 ± 0.253	7.147 ± 0.076
Dendrogram F1	0.034 ± 0.030	0.000 ± 0.000
Sibling recall@5	0.136 ± 0.003	0.188 ± 0.004
Cousin recall@5	0.249 ± 0.003	0.296 ± 0.005
Fréchet (Cubism) acc.	0.927 ± 0.029	0.913 ± 0.021

Table 1: Best Euclidean vs. best hyperbolic configuration across Phases 1–2, selected per geometry by mean top-1 accuracy across seeds. Mean $\pm$ seed standard deviation. Euclidean wins on every global metric except sibling and cousin recall, where hyperbolic retains a 4–5 percentage point lead.

5.4 Headline comparison

5.5 Qualitative analysis

6 Conclusions & Discussion

What we can claim. On a 27-style classification task built on frozen CLIP features, hyperbolic prototype embeddings produce *more* hierarchically faithful local neighborhoods than Euclidean ones (sibling and cousin recall@5 advantages of 4–5 pp), and the gap *widens* as the embedding dimension grows. They produce *less* hierarchically faithful global structure on classical metrics (top-1 accuracy, prototype-tree Spearman, dendrogram F1, mean tree distortion) at every scale tested. A hierarchy-aware regularizer improves global tree metrics in both geometries but does not flip the geometry winner on either local or global axes. These findings are stable across three seeds with standard deviations ≤ 1 pp on top-1 and ≤ 0.01 on the hierarchy metrics.

What we cannot claim. We cannot claim that hyperbolic embeddings are universally preferable for hierarchical image data — our results say the opposite for global metrics. We cannot claim that the hand-built tree is the "correct" style hierarchy; tree-Spearman and dendrogram F1 are anchored to that specific taxonomy. We cannot generalize to other domains: WikiArt-Refined is medium-scale (81k images), Western-canon-heavy, and the labels are themselves debated by art historians.

Limitations. Single dataset, single CLIP backbone, hand-built tree, and a single hierarchy-aware loss formulation (scale-invariant MSE on prototype distances). We did not explore hyperbolic versions of the head MLP itself (only the prototypes live on the manifold), per-style learnable curvature, or jointly learning the tree from the embeddings.

Future work. Three directions follow naturally. *(i)* Hyperbolic neural network layers throughout the head [2] rather than only at the prototype layer — this would let the manifold influence the representation rather than just the decision boundary. *(ii)* Learned trees from data [1], evaluated against the hand-built tree as one of several reference points, would test whether the WikiArt hand-built tree is a bottleneck. *(iii)* Extension to non-Western traditions where the lineage taxonomy

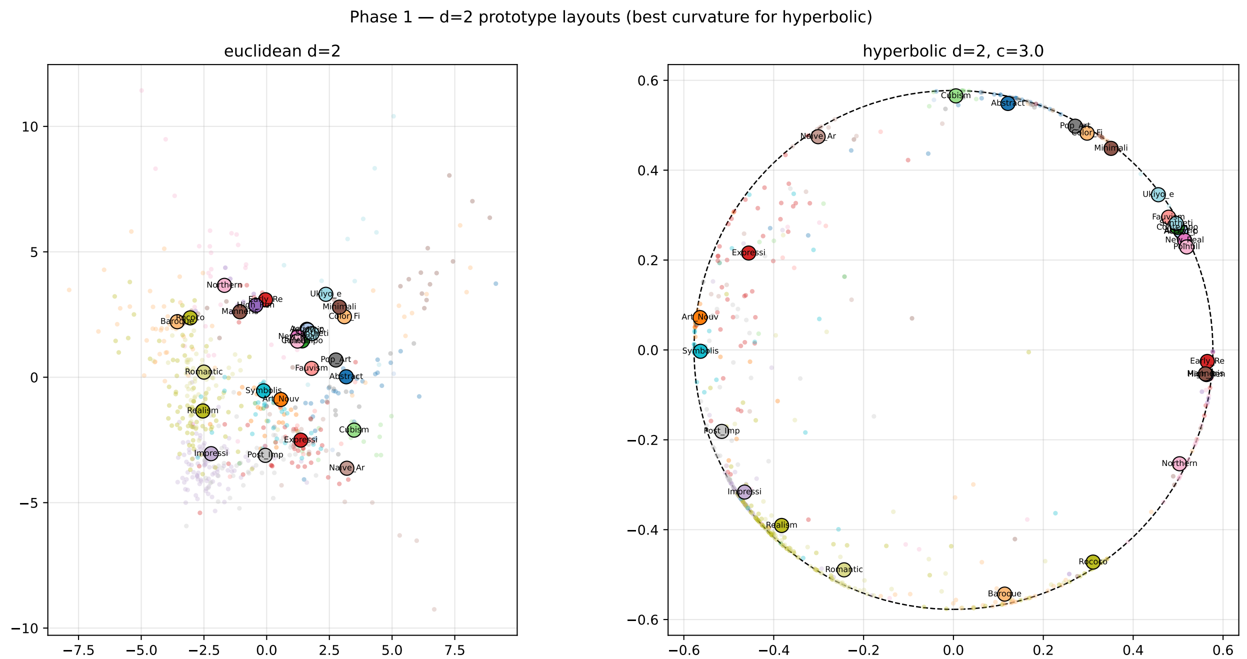

Figure 4: Prototype layouts at d=2 (best curvature per geometry). Each large dot is a learned style prototype; small dots are a sample of validation embeddings. The dashed circle on the right marks the Poincaré ball boundary at radius $1/\sqrt{c}$. Hyperbolic prototypes concentrate near the boundary — the exponential-volume regime — while Euclidean prototypes scatter without structural pressure. Theoretical prediction [5, 6] made visible on a real dataset.

underrepresents genuine structure (e.g. the East Asian ink-painting traditions collapsed into a single Ukiyo-e label here).

7 Broader Impact

The system we study is small (a 27-way style classifier on cached features), but the design choices have broader implications. Three points worth flagging.

First, our default ground-truth tree is built from a Western, lineage-based art-history canon. Distance metrics computed against it implicitly endorse that taxonomy. The Phase 3 ablation against alternative trees is partial mitigation; deployment in a museum or educational context should make the choice of tree explicit and revisable rather than hard-coded.

Second, the WikiArt corpus is biased toward European painting and sparsely represents non-Western traditions (Ukiyo-e is the only non-Western top-level category). A retrieval system trained on this data will under-rank non-Western works for ambiguous queries, and hyperbolic geometry — which we show preserves local hierarchy more faithfully — could amplify that effect by tightening neighborhoods inside the dominant taxonomy. Users of any downstream system should be informed of this bias.

Third, hyperbolic representations of style structure could in principle be used to attribute or authenticate paintings. We do not endorse such use without human-expert review: the classification accuracies reported here (∼50–65 % top-1 over 27 styles) are far below the threshold required for legal or commercial decisions.

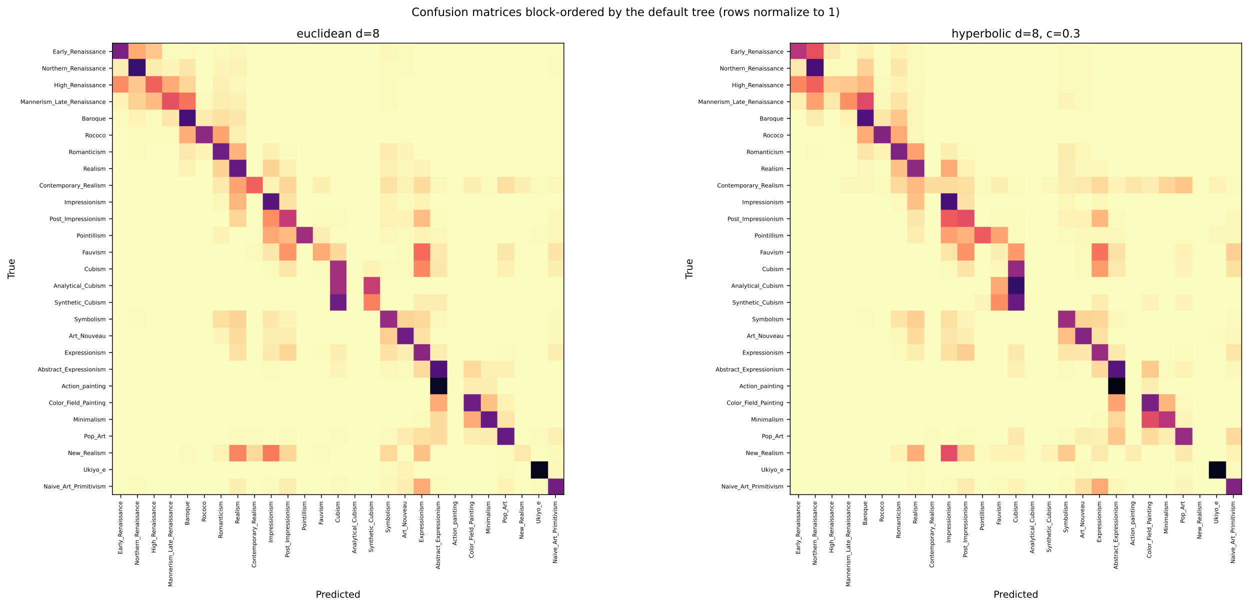

Figure 5: Row-normalized confusion matrices at d=8, rows and columns permuted by a depth-first traversal of the default style tree. Both models concentrate mistakes in near-diagonal blocks — errors predominantly fall on hierarchically adjacent styles. The hyperbolic block structure is visibly softer near the diagonal, consistent with its higher sibling and cousin recall.

References

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[4] Max Kochurov, Rasul Karimov, and Serge Kozlukov. Geoopt: Riemannian optimization in PyTorch. In *International Conference on Machine Learning, RIEML Workshop*, 2020.

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A Default style hierarchy

The default ground-truth tree, hand-built from standard art-history references and used as the primary evaluation reference throughout the paper. The complete definition lives in `scripts/hierarchy.py:STYLE_HIE`

```
Root
|-- Early_Renaissance
|   |-- Northern_Renaissance
|   `-- High_Renaissance
|       `-- Mannerism_Late_Renaissance
|           `-- Baroque
|               `-- Rococo
|                   `-- Romanticism
|                       |-- Realism
|                       |   |-- Contemporary_Realism
|                       |   `-- Impressionism
|                       |       `-- Post_Impressionism
|                       |           |-- Pointillism
|                       |           |-- Fauvism
|                       |           `-- Cubism
|                       |               |-- Analytical_Cubism
|                       |               `-- Synthetic_Cubism
|                       `-- Symbolism
|                           |-- Art_Nouveau
|                           `-- Expressionism
|                               `-- Abstract_Expressionism
|                                   |-- Action_painting
|                                   |-- Color_Field_Painting
|                                   `-- Minimalism
|-- Pop_Art
|   `-- New_Realism
|-- Ukiyo_e
`-- Naive_Art_Primitivism
```

B Sweep configuration

The sweep CSV (`notebooks/sweep_results.csv`, 150 rows) records, for every config, the full set of training hyperparameters and every evaluation metric. To reproduce:

```
python scripts/sweep.py --phase 1 --device mps   # ~14 min, 90 configs
python scripts/sweep.py --phase 2 --device mps   # ~10 min, 48 configs
python scripts/sweep.py --phase 3 --device mps   # ~3 min,  18 configs
python scripts/make_figures.py                   # regenerates docs/figures
```